Project MOnitoring

since our **voting app is running on AKS**, now we'll enable **Azure-native monitoring** the correct production way:

We want:

- ✅ Azure Monitor
- ✅ Kubernetes monitoring
- ✅ Compute / Cluster metrics
- ✅ Azure Managed Grafana dashboards

ARCHITECTURE (What we'll enable)

AKS → Azure Monitor (Container Insights) → Log Analytics → Managed Grafana → Dashboards

---

Enable Azure Monitor (Container Insights) on AKS

Option A (Portal – easiest)

1. Go to **Azure Portal**
2. Open your **AKS cluster**
3. Click **Insights**
4. Click **Enable**
5. Create or select a **Log Analytics Workspace**
6. Enable

That's it.

Option B (CLI)

Create Log Analytics workspace:

```
az monitor log-analytics workspace create \
  --resource-group rg-voting-app \
  --workspace-name aks-logs
```

Get ID:

```
az monitor log-analytics workspace show \
  --resource-group <rg> \
  --workspace-name aks-logs \
  --queryid -o tsv
```

You just need to register the provider.

Register Microsoft.Insights

Run:

```
az provider register --namespace Microsoft.Insights
```

Also register Log Analytics (just in case)

```
az provider register --namespace Microsoft.OperationalInsight
s
```

Verify Registration

Check status:

```
az provider show --namespace Microsoft.Insights --query regis
trationState
```

It should return:

```
"Registered"
```

If it says:

```
Registering
```

Wait 1–2 minutes and check again.

Run:

```
az monitor log-analytics workspace create --resource-group rg-voting-app --workspace-name aks-logs

WORKSPACE_ID=$(az monitor log-analytics workspace show --resource-group rg-voting-app --workspace-name aks-logs --query id -o tsv)

az aks enable-addons --resource-group rg-voting-app --name aks-voting --addons monitoring --workspace-resource-id $WORKSPACE_ID
```

✅ STEP 2 — Verify Metrics Are Flowing

After 5–10 minutes:

Go to:

```
AKS → Insights
```

You should see:

- Nodes
- Pods
- Containers
- CPU
- Memory
- Controllers

If you see data → Monitoring is working.

Create Azure Managed Grafana

Azure Managed Grafana integrates directly with Azure Monitor.

Create via CLI:

```
az grafana create \
  --name aks-grafana \
  --resource-group <rg>
```

OR use Portal:

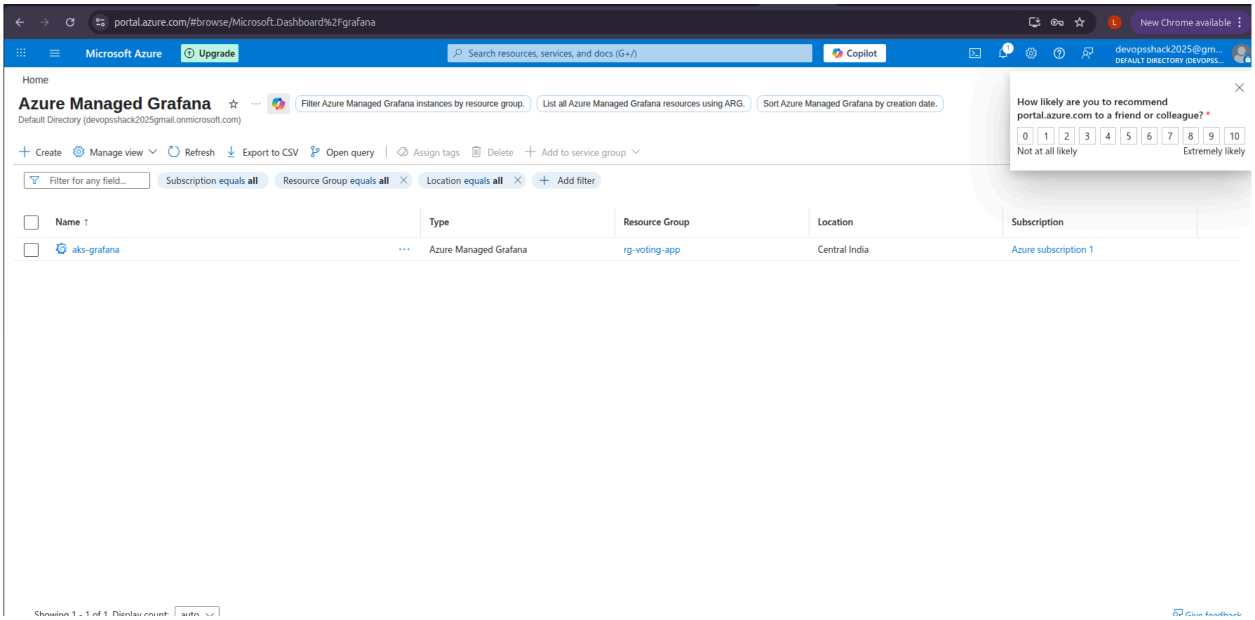
Create Resource → Search “Azure Managed Grafana”

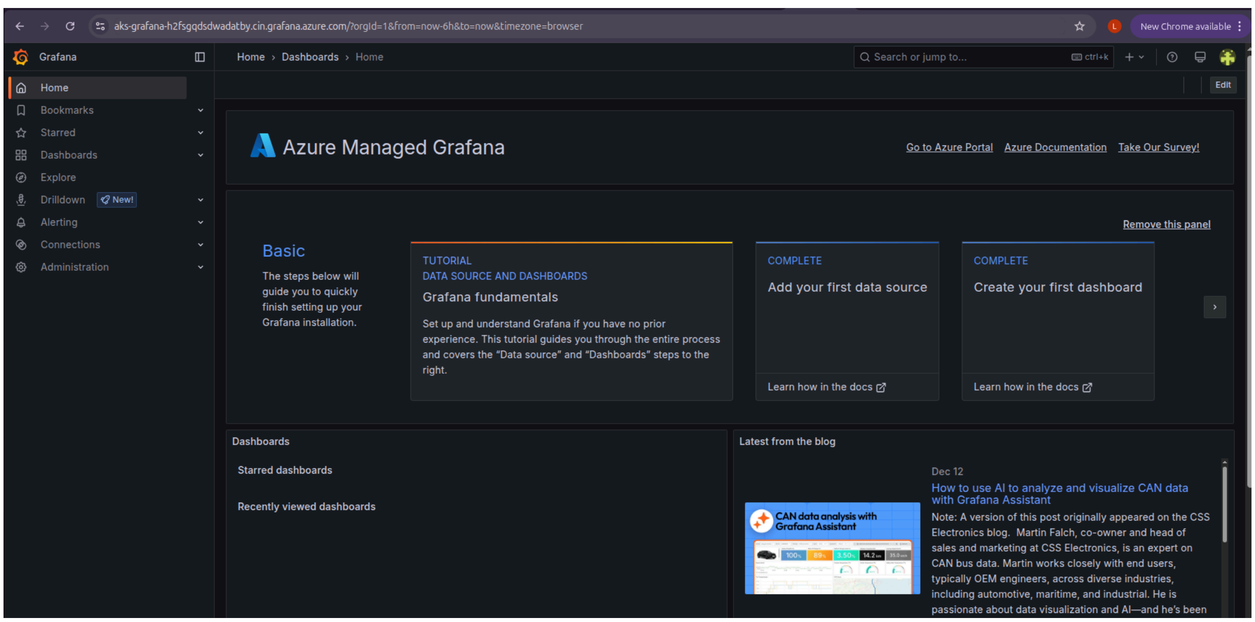
Connect Grafana to Azure Monitor

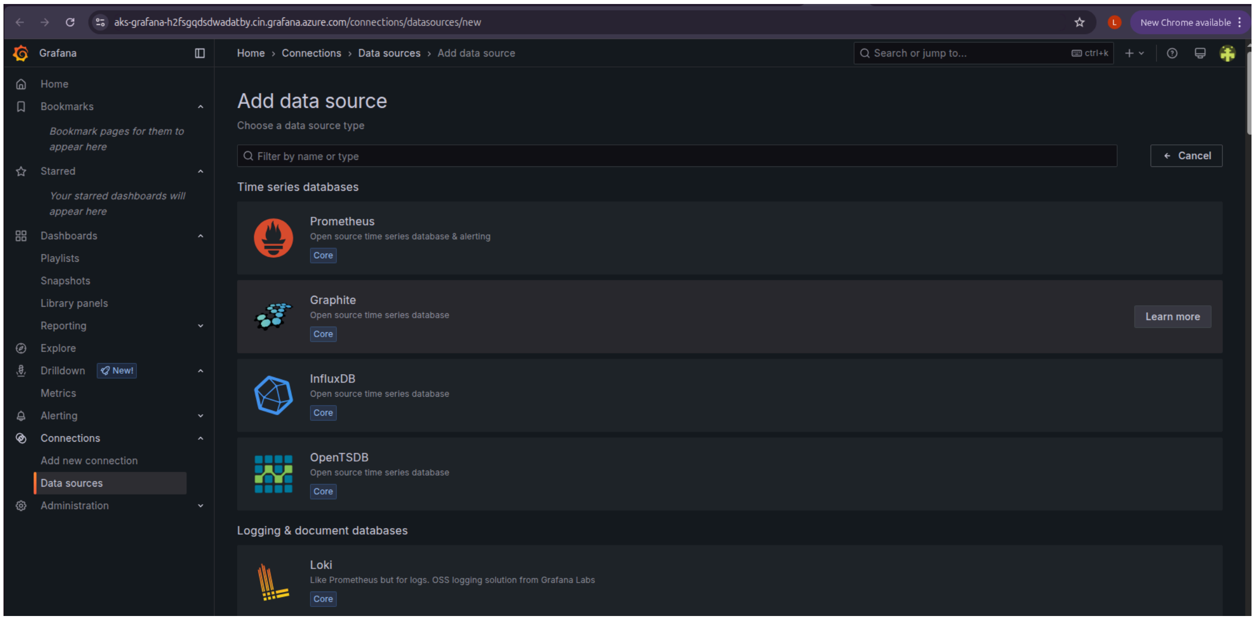
After Grafana is created:

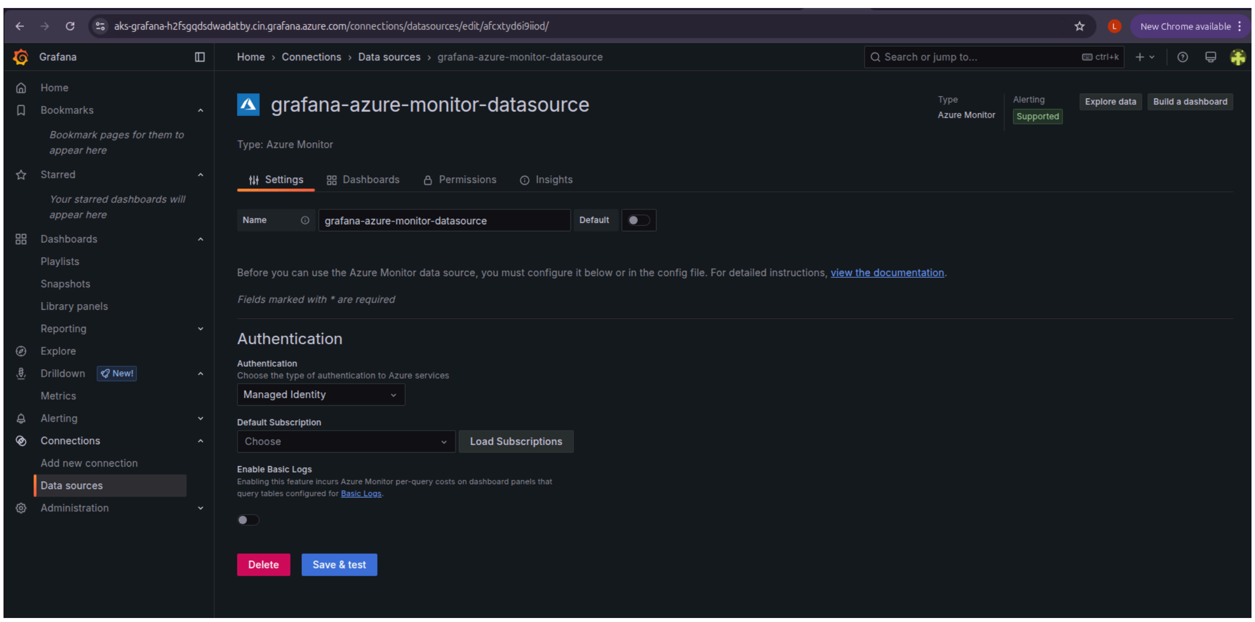
1. Open **Grafana resource**
2. Click **Endpoint**
3. Open it
4. Login using Azure AD

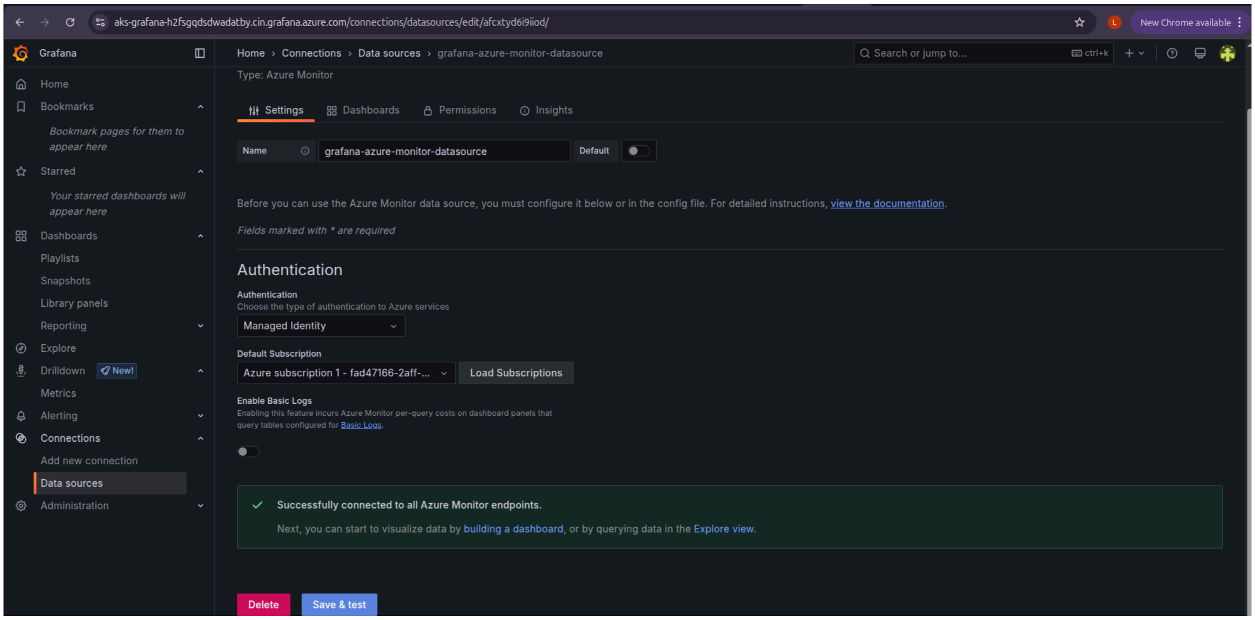
Now:


portal.azure.com/#browse/Microsoft.Dashboard%2Fgrafana
Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home
Azure Managed Grafana
Default Directory (devopsshack2025gmail.onmicrosoft.com)
Filter Azure Managed Grafana instances by resource group.
List all Azure Managed Grafana resources using ARG.
Sort Azure Managed Grafana by creation date.
How likely are you to recommend portal.azure.com to a friend or colleague?
Not at all likely
Extremely likely
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Delete
Add to service group
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Type
Resource Group
Location
Subscription
aks-grafana
Azure Managed Grafana
rg-voting-app
Central India
Azure subscription 1

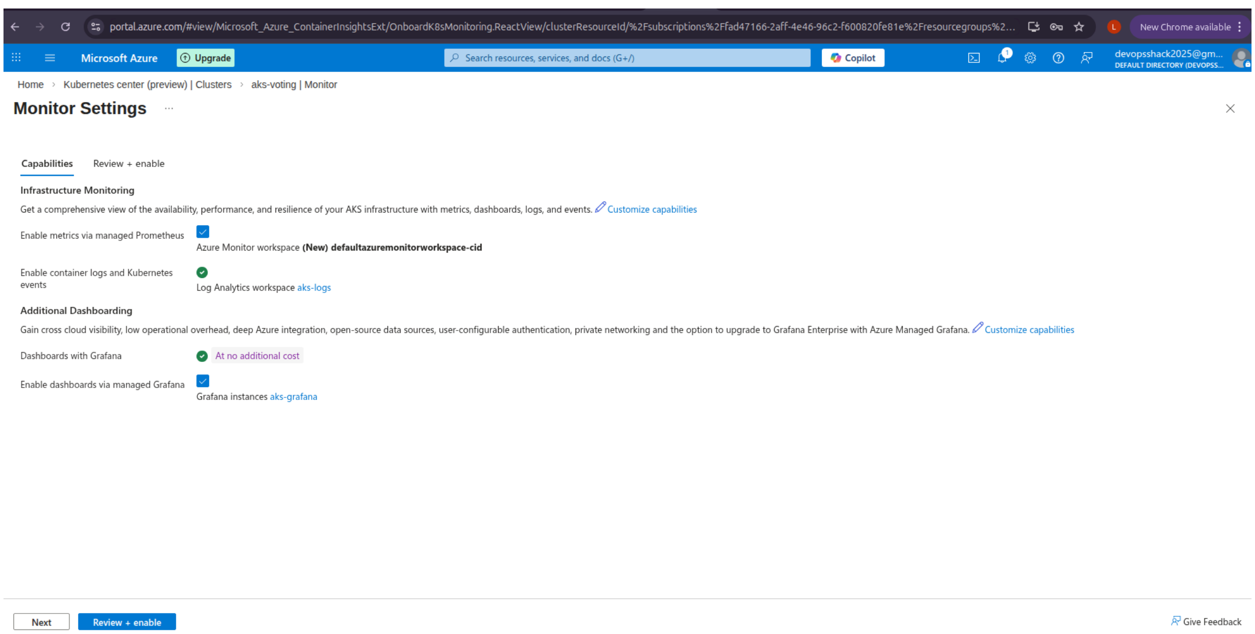

aks-grafana-h2fsgqdsdwadatby.cin.grafana.azure.com/?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Home › Dashboards › Home
Search or jump to...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
New!
Alerting
Connections
Administration
Edit
Azure Managed Grafana
Go to Azure Portal
Azure Documentation
Take Our Survey!
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
COMPLETE
Add your first data source
Learn how in the docs
COMPLETE
Create your first dashboard
Learn how in the docs
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
Dec 12
How to use AI to analyze and visualize CAN data with Grafana Assistant
CAN data analysis with Grafana Assistant
Note: A version of this post originally appeared on the CSS Electronics blog. Martin Falch, co-owner and head of sales and marketing at CSS Electronics, is an expert on CAN bus data. Martin works closely with end users, typically OEM engineers, across diverse industries, including automotive, maritime, and industrial. He is passionate about data visualization and AI—and he's been


aks-grafana-h2fsgqdsdwadatby.cin.grafana.azure.com/connections/datasources/new
New Chrome available
Grafana
Home › Connections › Data sources › Add data source
Search or jump to...
ctrl+k
Home
Bookmarks
Bookmark pages for them to appear here
Starred
Your starred dashboards will appear here
Dashboards
Playlists
Snapshots
Library panels
Reporting
Explore
Drilldown
New!
Metrics
Alerting
Connections
Add new connection
Data sources
Administration
Add data source
Choose a data source type
Filter by name or type
Cancel
Time series databases
Prometheus
Open source time series database & alerting
Core
Graphite
Open source time series database
Core
Learn more
InfluxDB
Open source time series database
Core
OpenTSDB
Open source time series database
Core
Logging & document databases
Loki
Like Prometheus but for logs. OSS logging solution from Grafana Labs
Core


aks-grafana-h2fsgqdsdwadatby.cin.grafana.azure.com/connections/datasources/edit/afcxtyd6i9iiod/
New Chrome available
Grafana
Home › Connections › Data sources › grafana-azure-monitor-datasource
Search or jump to...
ctrl+k
Home
Bookmarks
Bookmark pages for them to appear here
Starred
Your starred dashboards will appear here
Dashboards
Playlists
Snapshots
Library panels
Reporting
Explore
Drilldown
New!
Metrics
Alerting
Connections
Add new connection
Data sources
Administration
grafana-azure-monitor-datasource
Type
Azure Monitor
Alerting
Supported
Explore data
Build a dashboard
Type: Azure Monitor
Settings
Dashboards
Permissions
Insights
Name
grafana-azure-monitor-datasource
Default
Before you can use the Azure Monitor data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Authentication
Authentication
Choose the type of authentication to Azure services
Managed Identity
Default Subscription
Choose
Load Subscriptions
Enable Basic Logs
Enabling this feature incurs Azure Monitor per-query costs on dashboard panels that query tables configured for Basic Logs.
Delete
Save & test



Or use built-in Azure dashboards.



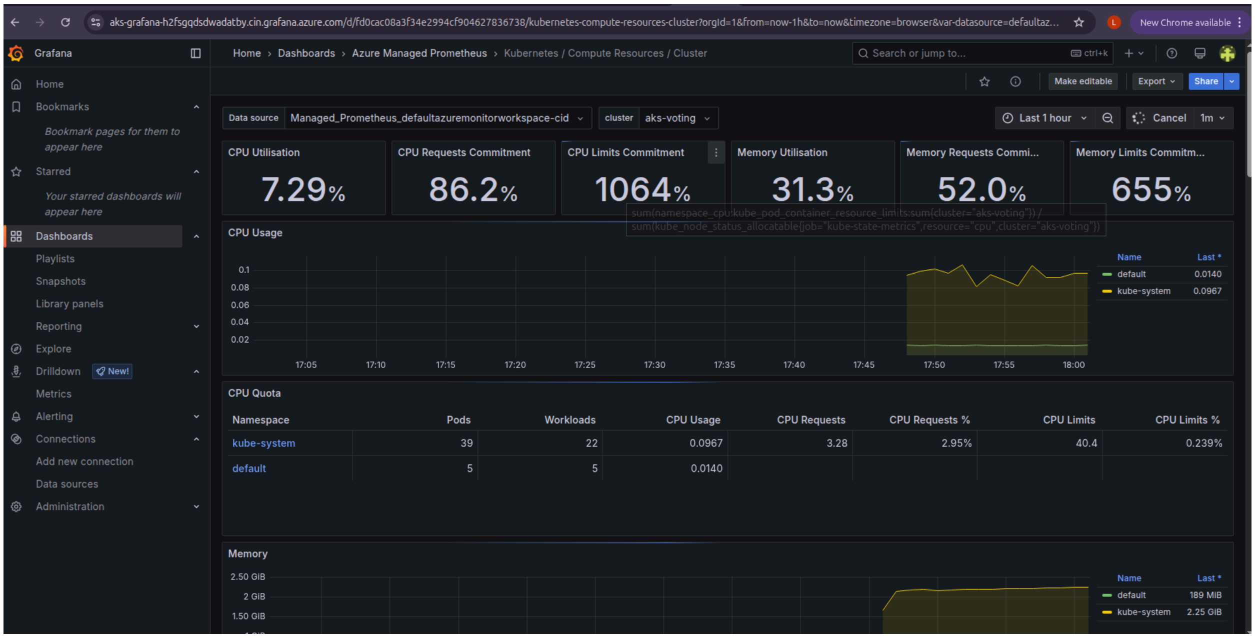
Click on the prometheus configure button on the monitor UI → Review+Create

Wait 5 minutes to get successful messege

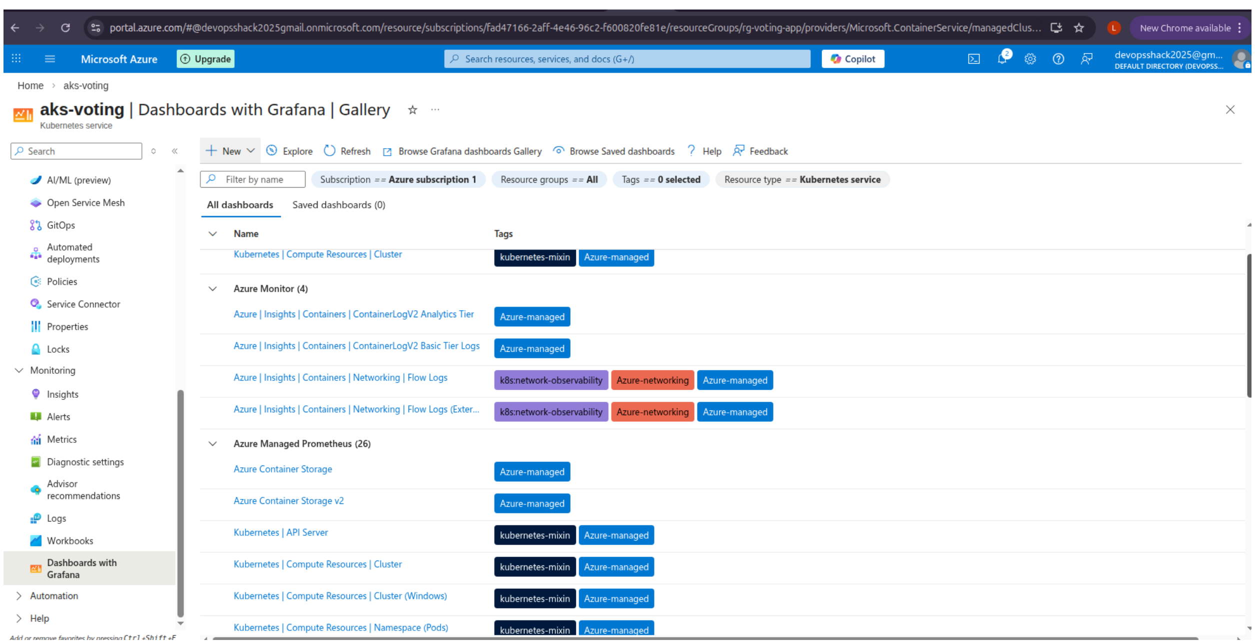
Then click on the grafana endpoint.

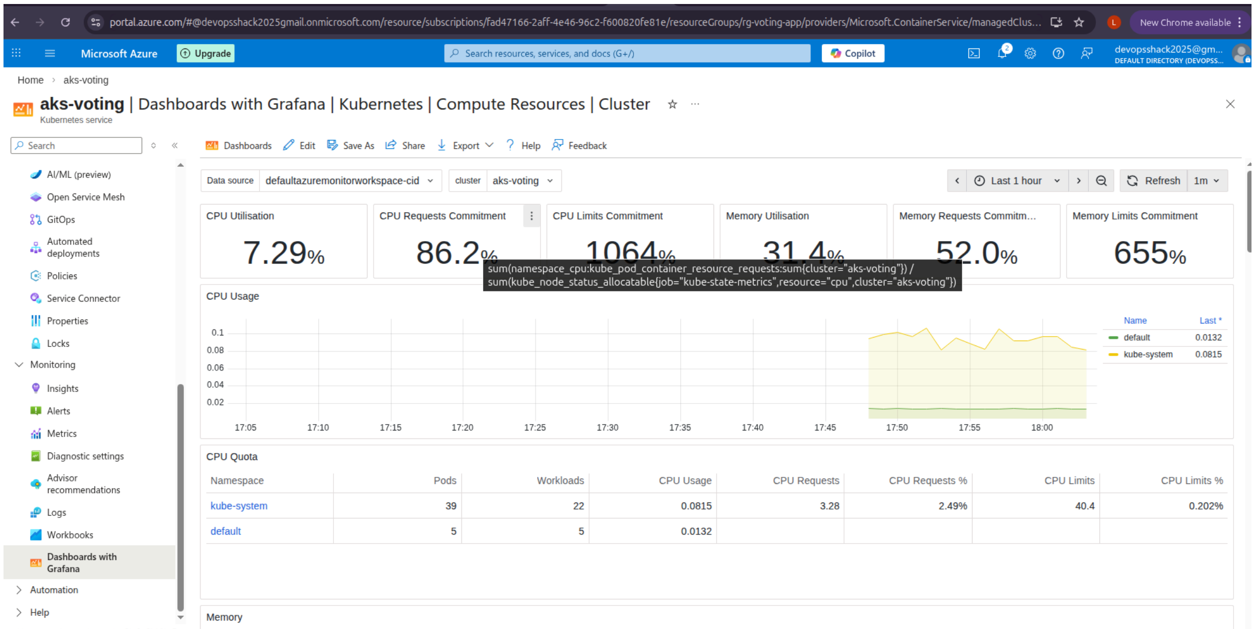
Navigate to Dashboard→ Azure managed prometheus.

See the dashboards:



Or you can head to Dasboard with grafana tab and view it directly form there.





Check the app pods